

BUILD GUIDE

Build Your Battle Bot

A simple guide to building a fighting robot you can control with your phone

What you're building

A small robot that drives around and spins a weapon to fight other robots. You control it with an Android phone via Bluetooth. Here's the basic idea: the robot has a brain (a tiny computer called the ESP32), two wheels (driven by small motors), and a spinning weapon (powered by a stronger motor). All of it runs on one rechargeable battery.

SAFETY FIRST

- **Always remove the weapon** before you power it on for testing. It spins automatically when the robot boots up.
- Never put your hands or face near the spinning parts when the battery is connected.
- The robot won't move or spin the weapon until you send a command from the app — but don't rely on that alone. Keep the battery off until you're ready.
- **Check battery polarity** (red and black wires) before plugging anything in. Getting this backwards will break the electronics instantly.
- When you're learning, keep the battery disconnected and turn it on only for testing.

Parts you need

PART	WHAT IT DOES	HOW MANY
ESP32-S3-Zero	The brain — tiny computer that talks to your phone	1
DRV8833	Motor controller for the drive wheels	1
N20 gear motor	Drives the wheels (6V rated)	2
Brushless motor (2204/2206)	Spins the weapon fast	1
Weapon ESC	Controller for the weapon motor (15A, has built-in 5V power supply)	1
2S LiPo battery	Power source (7.4V nominal)	1
Power switch	On/off switch rated for at least 15A	1
18AWG wire	For battery power connections (thick, for current)	varies
22AWG wire	For signal and motor connections (thin, for logic)	varies
Android phone	Your remote control via Bluetooth	1

1. Component overview

ESP32-S3-Zero (the brain)

A tiny computer with Bluetooth. It reads commands from your phone app and tells the motor controllers what to do. It gets power (5V) from the weapon ESC's built-in power supply, not directly from the battery.

Pins used: 5V (power in), GND (ground), IO6 (weapon signal), IO7 (left motor), IO8 (left motor), IO9 (right motor), IO10 (right motor)

DRV8833 (drive motor controller)

Lets the brain control the two wheel motors independently. Each wheel can go forward or backward. It needs power from the battery and signal wires from the brain.

Connections: VM (battery+ power), GND (ground), AIN1/AIN2 (left motor signals from brain), BIN1/BIN2 (right motor signals from brain), AOUT1/AOUT2 (to left wheel), BOUT1/BOUT2 (to

right wheel)

N20 gear motors (×2 for wheels)

Small 6V motors that drive the wheels. Each one gets two wires from the DRV8833. The two wires can swap to reverse direction — no polarity concern on first build.

2204/2206 brushless motor (weapon)

Spins very fast. Gets three wires from the weapon ESC. Order doesn't matter for direction on first build.

Weapon ESC (weapon controller)

Controls the brushless weapon motor AND provides 5V power to the brain via a special three-wire cable. The 5V wire powers the ESP32. It receives throttle signal from the brain's IO6.

2S LiPo battery (7.4V)

Red wire = positive (+), Black wire = negative (-). Powers everything through the switch.

2. Power wiring (battery to controllers)

Battery to switch

Red wire (positive) from battery connects to the switch. The switch will break and make this connection — it's your on/off.

Switch to power split

After the switch, the red wire splits into two paths:

- One path: **red wire** → **Weapon ESC "battery +" input**
- Second path: **red wire** → **DRV8833 "VM" input**

Battery negative to ground

Black wire (negative) from the battery goes to a common ground point. From there, connect to:

- DRV8833 GND input
- Weapon ESC "battery -" input
- ESP32 GND pin (though it also gets GND from the ESC's 3-wire cable)

✓ **Check:** Both the ESC and DRV8833 must have their ground (-) connected to the battery's black wire. The ESP32's ground is handled by the ESC's 3-wire cable, but you can also connect

it to the common ground.

3. ESC 5V power to brain

The weapon ESC has a three-wire cable that does two jobs:

- **Red wire (5V)** → ESP32 "5V" pin
- **Black wire (GND)** → ESP32 "GND" pin
- **Yellow/white wire (signal)** → ESP32 "IO6" pin

This three-wire cable provides both the power that runs the brain AND the command line that tells the ESC what throttle to apply. The brain's IO6 sends a servo-like pulse signal that the ESC interprets as weapon speed (0–100%).

✓ **Important:** The ESP32 is NOT wired to the battery directly. It gets all its power from the ESC's 5V output. This keeps the board safe and makes arming/disarming possible.

4. Drive motors to controller

Left wheel motor

Wire 1 of left N20 motor → DRV8833 **AOUT1**

Wire 2 of left N20 motor → DRV8833 **AOUT2**

Right wheel motor

Wire 1 of right N20 motor → DRV8833 **BOUT1**

Wire 2 of right N20 motor → DRV8833 **BOUT2**

The order of these two wires doesn't matter for a first build. If a wheel spins backwards during testing, just swap those two wires.

5. Weapon motor to ESC

The brushless motor has three wires (A, B, C). Connect them to the ESC's three motor output pads:

- Motor A → ESC output 1
- Motor B → ESC output 2
- Motor C → ESC output 3

The order doesn't matter for a first build. If the weapon spins backwards during testing, swap any two of the three motor wires.

6. Brain signals to drive controller

Thin signal wires from the ESP32 tell the DRV8833 what to do with each wheel:

Left wheel control (2 wires):

- ESP32 IO7 → DRV8833 AIN1
- ESP32 IO8 → DRV8833 AIN2

Right wheel control (2 wires):

- ESP32 IO9 → DRV8833 BIN2
- ESP32 IO10 → DRV8833 BIN1

✓ **Check:** These are signal wires (22AWG), not power wires. They tell the DRV8833 which direction and speed to spin each wheel based on what the app sends.

7. Building steps

1. **Mount the boards on the chassis.** Secure the ESP32-S3-Zero, DRV8833, and weapon ESC so they won't move. Keep wires organized so you can see what connects where.
2. **Wire the battery power.** Use 18AWG wire (thick, red and black). Battery red → switch → split to (ESC battery+) and (DRV8833 VM). Battery black → common ground → ESC battery-, DRV8833 GND, and ESP32 GND. Use solder joints, not tape.
3. **Wire the ESC's three-wire cable to the brain.** Red → ESP32 5V, Black → ESP32 GND, Yellow/white → ESP32 IO6. This both powers the brain and commands the weapon.
4. **Wire the left wheel motor to the DRV8833.** Left N20 wire 1 → AOUT1, wire 2 → AOUT2. Use 22AWG wire. Don't worry about polarity yet.
5. **Wire the right wheel motor to the DRV8833.** Right N20 wire 1 → BOUT1, wire 2 → BOUT2. Use 22AWG wire.
6. **Wire the weapon motor to the ESC.** Brushless motor A → ESC out 1, B → ESC out 2, C → ESC out 3. Use 22AWG wire. Order doesn't matter for first test.
7. **Wire the brain's control signals.** Use 22AWG wire:
 - ESP32 IO7 → DRV8833 AIN1

- ESP32 IO8 → DRV8833 AIN2
- ESP32 IO9 → DRV8833 BIN2
- ESP32 IO10 → DRV8833 BIN1

8. **Check all connections before powering up.** Inspect every wire end. Make sure red and black are right. Look for cold solder joints or exposed copper that could short. Remove the weapon blade/spinner.

8. Install Arduino IDE and set up the ESP32

Step 1: Download and install Arduino IDE

Windows:

1. Go to arduino.cc/en/software
2. Download the Windows Installer
3. Run the installer and follow the steps. Accept the default install location
4. Launch Arduino IDE when done

macOS:

1. Go to arduino.cc/en/software
2. Download the macOS version (choose Intel or Apple Silicon depending on your Mac)
3. Open the .zip file and drag Arduino.app to your Applications folder
4. Launch from Applications

Linux:

1. Go to arduino.cc/en/software
2. Download the Linux version for your architecture (x86_64 or ARM64)
3. Extract the .tar.xz file: `tar -xf arduino-*.tar.xz`
4. Navigate into the extracted folder: `cd arduino-*`
5. Some versions include `install.sh` — if present, run it. Otherwise, directly launch: `./arduino`
6. You may need to add yourself to the dialout group to access USB ports: `sudo usermod -a -G dialout $USER` then log out and back in

Step 2: Add the ESP32 board package

1. Open Arduino IDE
2. Go to **File** → **Preferences** (Arduino → Preferences on Mac)
3. Find the field called "Additional Boards Manager URLs"
4. Paste this link into the field: `https://raw.githubusercontent.com/espressif/arduino-esp32/main-h-pages/package_esp32_index.json`
5. Click OK
6. Go to **Tools** → **Board** → **Boards Manager**

7. Search for "ESP32" by Espressif Systems
8. Click Install (this downloads the board definitions)
9. Wait for it to finish — it's about 500 MB, may take a few minutes

Step 3: Install the ESP32Servo library

1. Go to **Sketch** → **Include Library** → **Manage Libraries**
2. Search for "ESP32Servo"
3. Find the one by Kevin Harrington, click Install

Step 4: Connect your board and get the code

1. Plug your ESP32-S3-Zero into your computer via USB-C
2. Go to github.com/thefuzzman/WSP_YM_BB to download the bot's sketch
3. Click the green "Code" button and select "Download ZIP"
4. Extract the ZIP file and open the .ino file in Arduino IDE

Step 5: Configure the code

Before uploading, set a unique name and password for your bot. Find these lines near the top of the code:

```
const char* BOT_NAME = "MyBot";  
const char* BOT_PIN = "1234";
```

Change "MyBot" to a name you want (keep it under 15 characters, no spaces). Change "1234" to a 4-digit PIN only you'll know.

Step 6: Select the board and upload

1. Go to **Tools** → **Board** → **ESP32S3 Dev Module** (this is the WaveShare ESP32-S3-Zero board)
2. Go to **Tools** → **Port** and select the COM port (Windows, e.g. COM3) or /dev/cu.* (Mac) or /dev/ttyUSB* (Linux) that appeared when you plugged in the board
3. Click the Upload button (arrow icon in the top left) or press Ctrl+U (Cmd+U on Mac)
4. Wait for "Done uploading" message at the bottom — the board will reboot automatically

9. Using the Android app

Initial Bluetooth permission

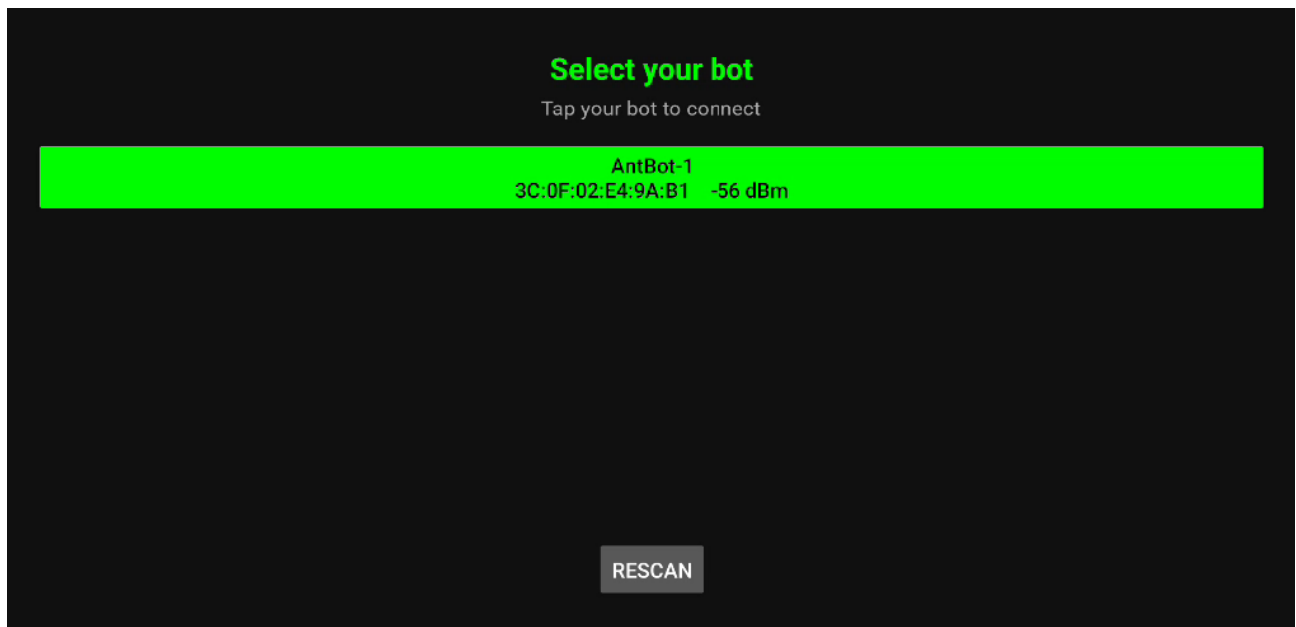
When you launch the app for the first time, it asks for permission to use Bluetooth. Tap **Allow** so the app can find and connect to your bot.



The app asks for Bluetooth permission on first launch

Finding and selecting your bot

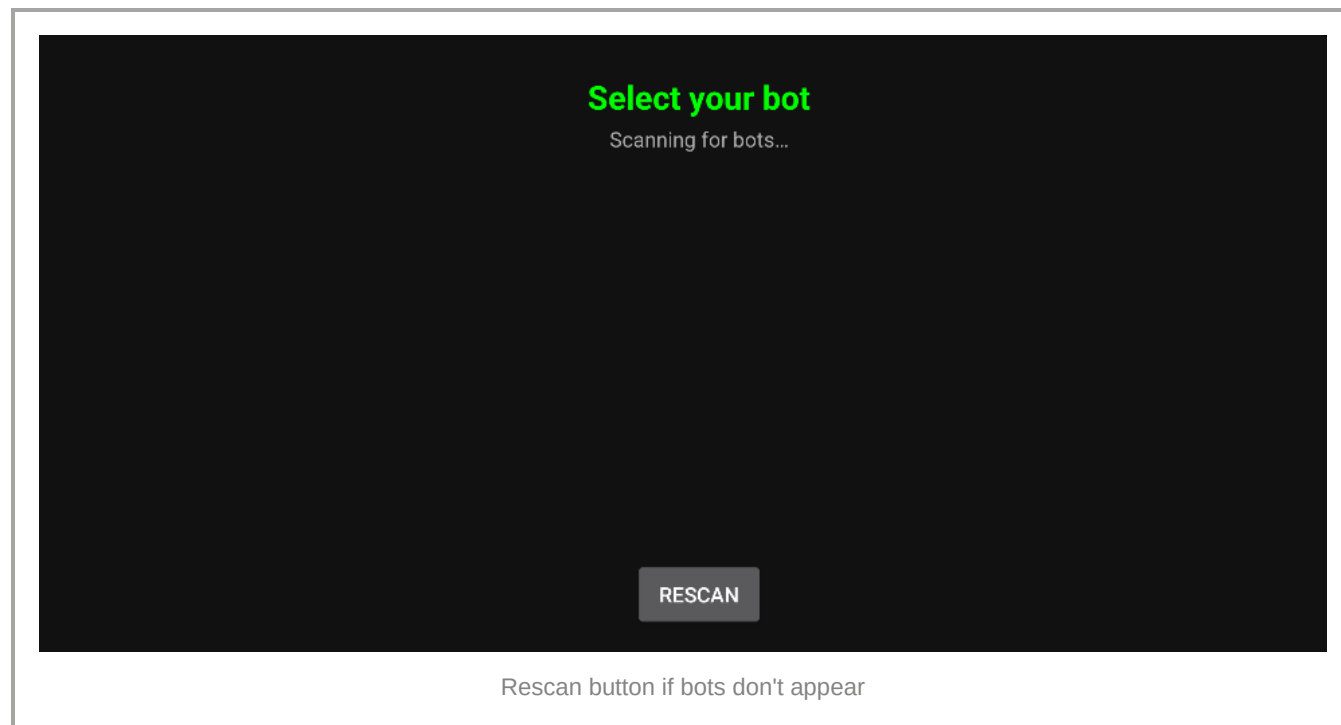
The app scans for nearby bots and displays them by the name you set in the code (BOT_NAME). Tap the bot's name to connect.



Bot selection screen showing discovered bots

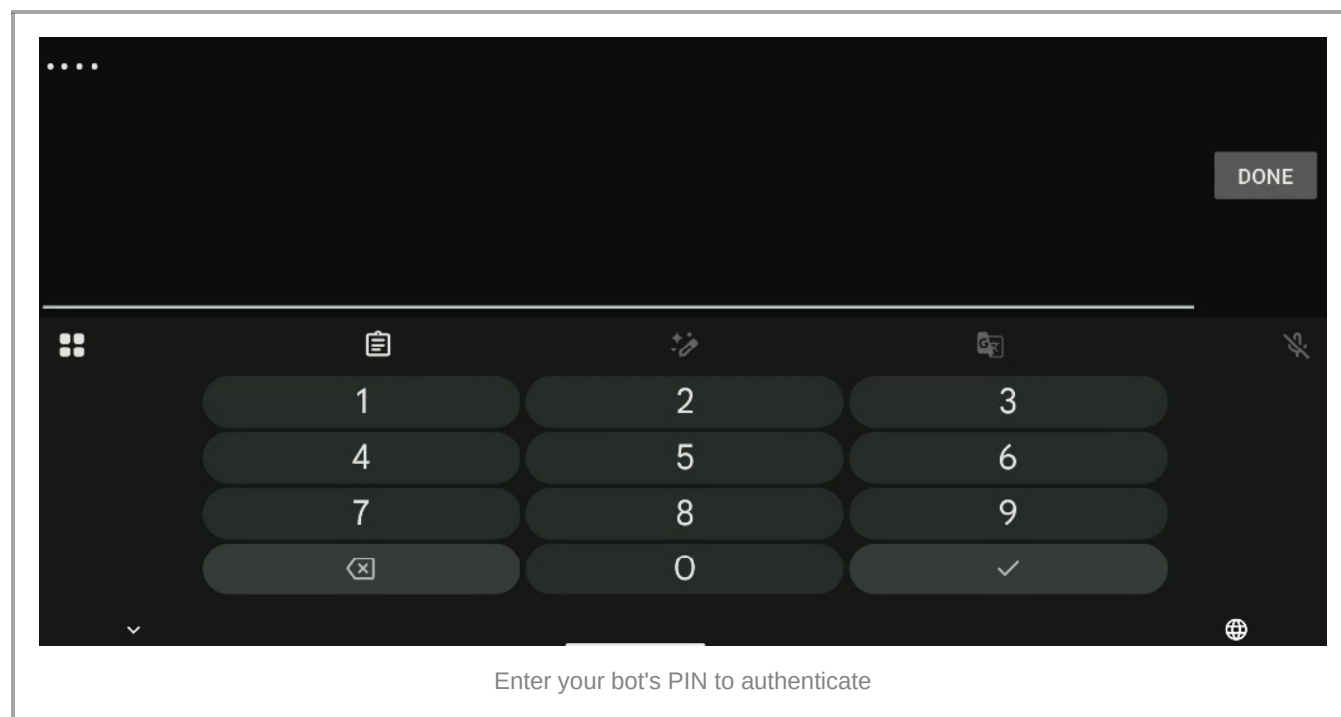
If your bot doesn't appear: Make sure the bot is powered on (switch flipped) and the ESP32's LED

is lit. If it still doesn't show, tap **RESCAN** to search again.



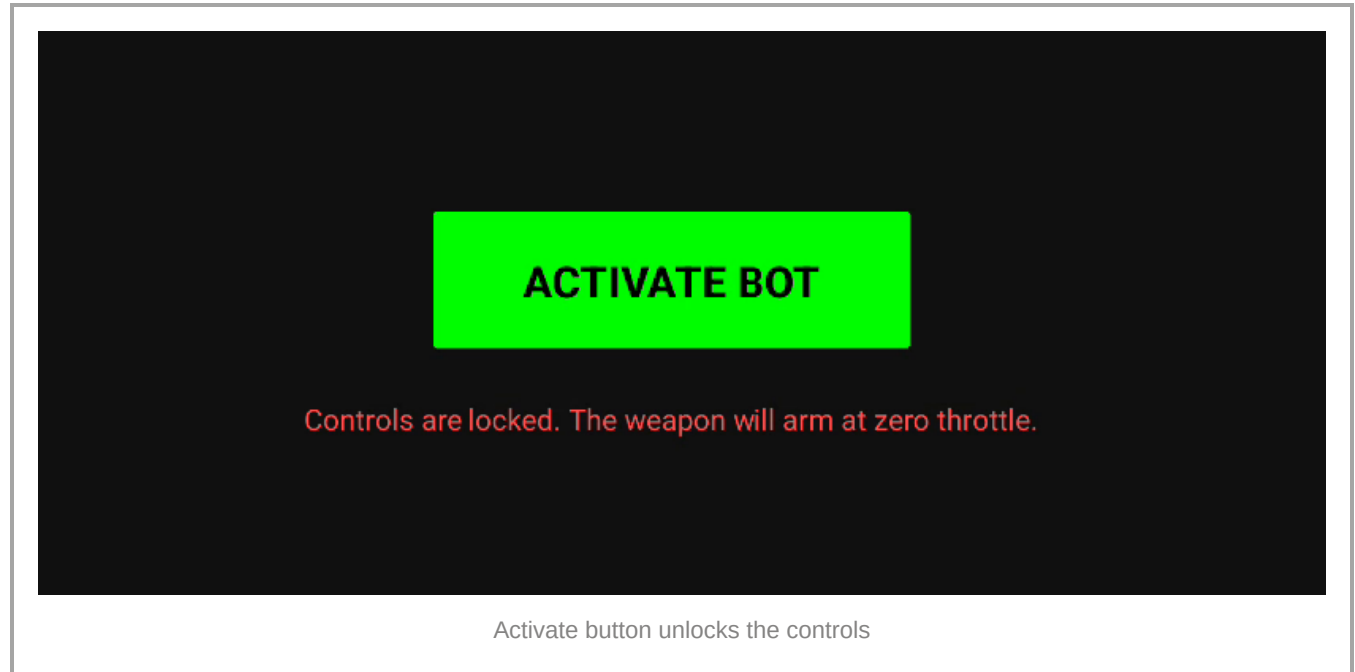
Entering the PIN

Once you select your bot, the app asks for the PIN you set in the code (BOT_PIN). Type the 4-digit PIN and tap the checkmark.



Activating the bot

After authentication, the app shows an **ACTIVATE BOT** button. The message says "Controls are locked. The weapon will arm at zero throttle." This is a safety feature — tap **ACTIVATE BOT** to unlock the controls and arm the weapon motor.



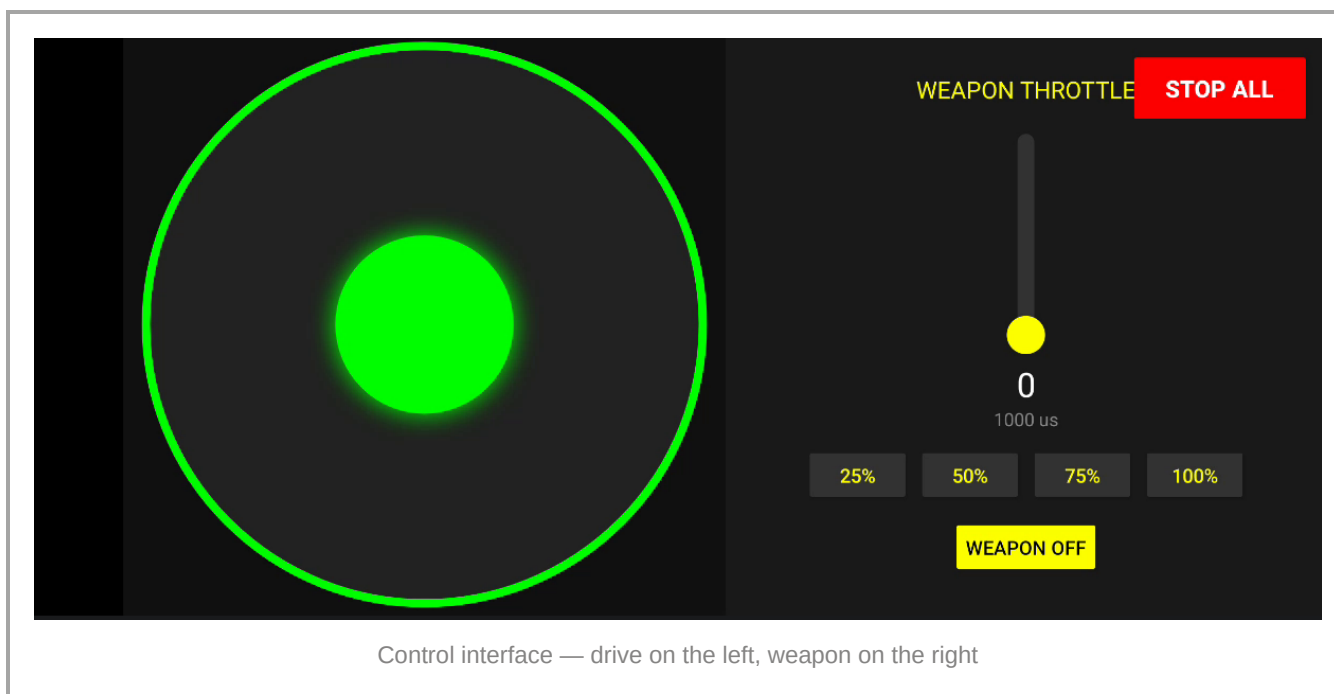
Control interface

Left side (green circle): This is the driving joystick. Drag your finger to control forward/backward and left/right turns. Center = stopped.

Right side (yellow dot): This is the weapon throttle. Drag up to increase weapon speed, down to decrease. Four preset buttons (25%, 50%, 75%, 100%) let you lock to a specific power level.

WEAPON OFF button: Instantly kills the weapon motor (sets it to 0%).

STOP ALL button (red): Emergency stop — kills everything immediately. Use this if something goes wrong.



10. First power-on

1. **Weapon removed.** Check this twice.
2. **Flip the power switch on.** Watch the ESP32 board for a light (LED). If it lights up (purple, blue, or any color), the brain is powered and the ESC's power supply is working.
3. **Wait 3 seconds.** The ESC needs time to arm itself. You might hear a beep from the weapon controller.
4. **Open the Android app.** Scan for your bot's name. If it doesn't show up, check that the ESP32's LED is actually on. If the LED is off, something is wrong with power.
5. **Connect and enter the PIN.** The app asks for the PIN you set in the code. Type it in. The app should confirm "Connected".
6. **Tap "Activate" in the app.** This tells the brain it's okay to move. The light on the brain should change color (usually green).
7. **Test the drive controls.** Gently move the left/right drive sliders in the app. Both wheels should start spinning. If one wheel spins the wrong way, turn off the switch, then swap those two wires at the DRV8833.
8. **Test the weapon (gently).** With the weapon slider, slowly increase throttle. The motor should spin faster. If it's backward, swap any two of the three ESC-to-motor wires and re-test.
9. **Hit STOP.** The app's STOP button should kill the motors instantly. If it does, power is controlled correctly.

Common issues & fixes

No light on the ESP32 when you flip the switch:

- Check battery polarity — red should be positive, black negative
- Test the power switch with a multimeter to make sure it's passing power
- Check solder joints on the battery and ESC connections — they must be solid

ESP32 lights up but app doesn't see the bot:

- Make sure Bluetooth is on on your phone
- Make sure the BOT_NAME in the code matches what you're searching for in the app
- Try uploading the code again

Wheels spin backwards or not at all:

- Check that the signal wires from the ESP32 (IO7, IO8, IO9, IO10) are connected to the right DRV8833 pins
- If spinning backwards, swap the two wires for that wheel at the DRV8833

Weapon doesn't spin or spins very slowly:

- Check that IO6 from the ESP32 is connected to the ESC's signal wire (yellow/white)
- Make sure the three motor wires are solid at the ESC output pads
- If spins the wrong direction, swap any two of the three motor wires

Ready to fight?

- Install the weapon blade or spinner
- Fully charge the battery
- Check the power switch works (on and off)
- Test the STOP button in the app — it must kill power immediately
- Turn on via app, drive around for 10 seconds to confirm both wheels respond
- Hit STOP, turn off the switch, remove the weapon, and you're ready